

Project Initialization and Planning Phase

Date	14 July 2026
Team ID	
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

OptiCrop – Smart Agricultural Production Optimization Engine uses machine learning to analyze soil nutrients and environmental conditions and recommend the most suitable crop. The goal is to reduce guesswork in crop selection, improve productivity, and support sustainable, data-driven farming decisions.

I am	I'm trying to	But	Because	Which makes me feel
(Farmer)	(select the right crop for my land)	(soil and climate conditions keep changing)	(traditional experience-based selection)	(uncertain and worried)

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer selecting a crop to cultivate.	Choose the crop best suited to my soil and climate.	Crop choice today relies mostly on past experience and guesswork.	Soil nutrients, temperature, humidity, pH, and rainfall must all be balanced for a crop to succeed.	Uncertain and worried about crop failure and financial loss.